BABU BANARSI DAS UNIVERSITY

NO SQL and Dbaas

(BCADSN13202)

PROJECT

SUBMITTED TO:

Mr Ankit Verma

SUBMITTED BY:

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BCADS25

Q1. List the names and departments of students who have more than 85% attendance and are skilled in both "MongoDB" and "Python".

Solution: - db.students.find({attendance: { $gt: 85 },skills: { $all: ["MongoDB", "Python"] }},{_id: 0,name: 1,department: 1})

Explanation: -

• attendance: { **$gt**: 85 } → means attendance greater than 85.

• **$all** → checks that both skills “MongoDB” and “Python” are present.

• Projection { name: 1, department: 1 } → shows only name and department.

Output: No record found.

```
db> db.students.find({attendance: { $gt: 85 },skills: { $all: ["MongoDB", "Python"] }},{_id: 0,name: 1,department: 1}) //Pratistha Srivastava 1240258328

db>
```

Q2. Show all faculty who are teaching more than 2 courses. Display their names and the total number of courses they teach.

Solution: - db.faculty.aggregate([{ $project: { name: 1, number_of_courses: { $size: "$courses" } } }, { $match: { number_of_courses: { $gt: 2 } } }])

Explanation: -

• **$size** → counts how many items are in the “courses” array.

• **$match** → filters only those who teach more than 2 courses.

Output:

```
db> db.faculty.aggregate([ { $project: { name: 1, number_of_courses: { $size: "$courses" } } }, { $match: { number_of_courses: { $gt: 2 } } } ]) // Pratisth
a Srivastava 1240258328
[
  { _id: 'F029', name: 'Charles Newton', number_of_courses: 3 },
  { _id: 'F032', name: 'Julia Cole', number_of_courses: 3 },
  { _id: 'F040', name: 'Darrell Velasquez', number_of_courses: 3 },
  { _id: 'F048', name: 'Michael Poole', number_of_courses: 3 },
  { _id: 'F051', name: 'John Duran', number_of_courses: 3 },
  { _id: 'F061', name: 'Daniel Allen', number_of_courses: 3 },
  { _id: 'F083', name: 'Matthew Hanna', number_of_courses: 3 },
  { _id: 'F084', name: 'Michael Johnson', number_of_courses: 3 },
  { _id: 'F100', name: 'Robert Lara', number_of_courses: 3 }
]
```

Q3. Write a query to show each student's name along with the course titles they are enrolled in (use $lookup between enrollments, students, and courses).

Solution: - db.enrollment.aggregate([{$lookup: {from: "students",localField: "student_id",foreignField: "_id",as: "student"}},{ $unwind: "$student" },{$lookup: {from: "courses",localField: "course_id",foreignField: "_id",as: "course"}},{ $unwind: "$course" },{$project: {_id: 0,student_name: "$student.name",course_title: "$course.title"}}])

Explanation: -

- **$lookup →** joins collections (like SQL JOIN).
- **$unwind →** flattens joined results.
- **$project** → shows only student name and course title.

```
db> db.enrollment.aggregate([{$lookup: {from: "students",localField: "student_id",foreignField: "_id",as: "student"}},{ $unwind: "$student" },{$lookup: {fro
m: "courses",localField: "course_id",foreignField: "_id",as: "course"}},{ $unwind: "$course" },{$project: {_id: 0,student_name: "$student.name",course_ti
tle: "$course.title"}}])  // Pratishtha Srivastava 1240258328
[
  {
    student_name: 'Alexandra Bailey',
    course_title: 'Reactive neutral adapter'
  },
  {
    student_name: 'Megan Taylor',
    course_title: 'Sharable bifurcated paradigm'
  },
  {
    student_name: 'Alejandro Hart',
    course_title: 'Focused user-facing paradigm'
  },
  {
    student_name: 'Timothy Sparks',
```

Q4. For each course, display the course title, number of students enrolled, and average marks (use $group).

Solution: - db.enrollment.aggregate([{ $lookup: { from: "courses", localField: "course_id", foreignField: "_id", as: "course" } }, { $unwind: "$course" }, { $group: { _id: "$course_id", course_title: { $first: "$course.title" }, total_students: { $sum: 1 }, average_marks: { $avg: "$marks" } } }])

Explanation: -

- **$lookup →** connects enrollments with courses.
- **$group →** groups all students by course.
- **$sum: 1 →** counts students.
- **$avg →** calculates average marks.

```
db> db.enrollment.aggregate([ { $lookup: { from: "courses", localField: "course_id", foreignField: "_id", as: "course" } }, { $unwind: "$course" }, { $group
: { _id: "$course_id", course_title: { $first: "$course.title" }, total_students: { $sum: 1 }, average_marks: { $avg: "$marks" } } } ]) // Pratishtha Srivas
tava 1240258328
[
  {
    _id: 'C090',
    course_title: 'Configurable scalable data-warehouse',
    total_students: 2,
    average_marks: 77.5
  },
  {
    _id: 'C013',
    course_title: 'Enhanced radical secured line',
    total_students: 1,
    average_marks: 51
  },
```

Q5. Find the top 3 students with the highest average marks across all enrolled courses.

Solution: db.enrollment.aggregate([{ $group: { _id: "$student_id", average_marks: { $avg: "$marks" } } }, { $lookup: { from: "students", localField: "_id", foreignField: "_id", as: "student" } }, { $unwind: "$student" }, { $project: { _id: 0, student_name: "$student.name", average_marks: 1 } }, { $sort: { average_marks: -1 } }, { $limit: 3 }])

Explanation: -

- **$group →** Groups by student_id and calculates the average marks.
- **$lookup →** Joins with students_full to get student names.
- **$sort →** Orders by average_marks in descending order.
- **$limit** → Shows only top 3 students.

Output: -

```
db> db.enrollment.aggregate([{ $group: { _id: "$student_id", average_marks: { $avg: "$marks" } } }, { $lookup: { from: "students", localField: "_id", foreignField: "_id", as: "student" } }, { $unwind: "$student" }, { $project: { _id: 0, student_name: "$student.name", average_marks: 1 } }, { $sort: { average_marks: -1 } }, { $limit: 3 }]) //Pratishtha Srivasatava 1240258328
[
  { average_marks: 100, student_name: 'Diane Phillips' },
  { average_marks: 98, student_name: 'Brandon Rios' },
  { average_marks: 94, student_name: 'Larry Ramsey' }
]
db>
```

Q6. Count how many students are in each department. Display the department with the highest number of students.

Solution: - db.students.aggregate([{ $group: { _id: "$department", total_students: { $sum: 1 } } }, { $sort: { total_students: -1 } }, { $limit: 1 }])

Explanation: -

- **$group →** Groups students by department and counts them.
- **$sort →** Orders by student count in descending order.
- **$limit: 1 →** Shows only the department with the highest number of students.

Output: -

```
db> db.students.aggregate([ { $group: { _id: "$department", total_students: { $sum: 1 } } }, { $sort: { total_students: -1 } }, { $limit: 1 } ])
 // Pratishtha srivasatava 1240258328
[ { _id: 'Electrical', total_students: 23 } ]
db>
```

Q7. Update attendance to 100% for all students who won any "Hackathon".

Solution: - db.students.updateMany({activities : "Hackathon "}, { $set : { attendance : 100 }})

Explanation: -

- **$set** - is used to add a new field or update an existing field in documents.
- **updateMany() -**is used to update multiple documents in a collection that match a given condition.

Output:-

```
db> db.students.updateMany({activities : "Hackathon "}, { $set : { attendance : 100 }}) //Pratishtha Srivastava 1240258328
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Q8. Delete all student activity records where the activity year is before 2022.

Solution: - db.activities.deleteMany({ year: { $lt: 2022 } })

Explanation: -

- **$lt: 2022 →** Finds activities before 2022**(less than 2022)**
- **deleteMany →** Removes all matching documents

```
db>  db.activities_full.deleteMany({ year: { $lt: 2022 } })
{ acknowledged: true, deletedCount: 0 }
db>
```

Q9. Upsert a course record for "Data Structures" with ID "C150" and credits 4—if it doesn't exist, insert it; otherwise update its title to "Advanced Data Structures".

Solution: - db.courses_full.updateOne({ _id: "C150" },{ $set: { title: "Advanced Data Structures", credits: 4 } },{ upsert: true })

Explanation: -

- **updateOne →** Updates a single document.
- **$set →** Updates title and credits.
- **upsert: true →** If _id: "C150" doesn't exist, it will insert a new document.

Output:-

```
db> db.courses_full.updateOne({ _id: "C150" },{ $set: { title: "Advanced Data Structures", credits: 4 } },{ upsert: true }) //Pratishtha Srivastava 12402583
28
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Q10. Find all students who have "Python" as a skill but not "C++".

Solution: - db.students.find({skills: "Python",skills: { $ne: "C++" }},{ _id: 0,name: 1,skills: 1})

Explanation: -

- **skills: "Python" →** Checks that the array includes "Python".
- **skills: { $ne: "C++" } →** Ensures "C++" is not in the array.
- **Projection { name: 1, skills: 1 } →** Shows only names and skills.

Output: -

```
db> db.students.find({skills: "Python",skills: { $ne: "C++" }},{ _id: 0,name: 1,skills: 1}) //Pratishtha Srivastava 1240258328
[
  { name: 'Bruce Blair', skills: [ 'MongoDB', 'Linux' ] },
  { name: 'Alexandra Bailey', skills: [ 'Research', 'AutoCAD' ] },
  { name: 'Kyle Hale', skills: [ 'Python', 'Java' ] },
  { name: 'Daniel Robinson', skills: [ 'JavaScript', 'Java' ] },
  { name: 'Tina Hodge', skills: [ 'SQL', 'Research' ] },
  { name: 'Anthony Zavala', skills: [ 'Java', 'Git' ] },
  { name: 'Cody Whitehead', skills: [ 'JavaScript', 'Python' ] },
  { name: 'Thomas Jackson', skills: [ 'Python', 'AutoCAD' ] },
  { name: 'Monica Martin', skills: [ 'Research', 'JavaScript' ] },
  { name: 'Kathryn Ferguson', skills: [ 'Java', 'Linux' ] },
  { name: 'Steven Wong', skills: [ 'MongoDB', 'Python' ] },
  { name: 'Daniel Brown', skills: [ 'MongoDB', 'Research' ] },
  { name: 'Jason Brown', skills: [ 'MongoDB', 'SQL' ] },
  { name: 'Cheryl Jackson', skills: [ 'Research', 'Python' ] },
```

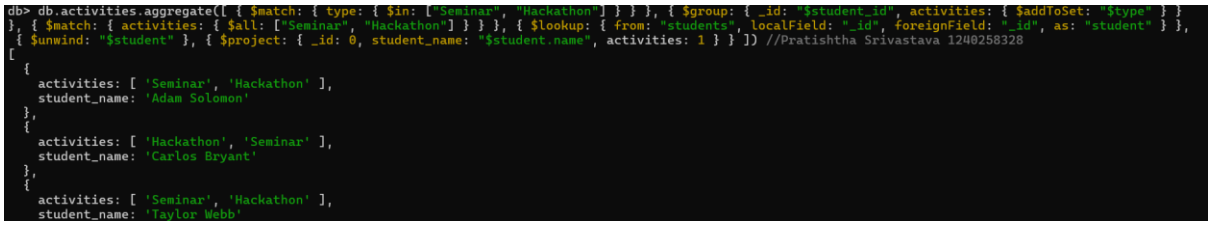
Q11. Return names of students who participated in "Seminar" and "Hackathon" both.

Solution: - db.activities.aggregate([{ $match: { type: { $in: ["Seminar", "Hackathon"] } } }, { $group: { _id: "$student_id", activities: { $addToSet: "$type" } } }, { $match: { activities: { $all: ["Seminar", "Hackathon"] } } }, { $lookup: { from: "students", localField: "_id", foreignField: "_id", as: "student" } }, { $unwind: "$student" }, { $project: { _id: 0, student_name: "$student.name", activities: 1 } }])

Explanation: -

- **$group →** Groups by student and collects unique activity types.
- **$match →** Keeps only students who have both activities.
- **$lookup →** Gets student names from students.
- **$project →** Shows only student name and their activities.

Output:



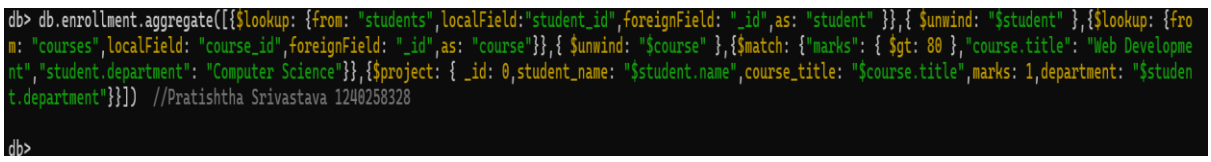
Q12. Find students who scored more than 80 in "Web Development" only if they belong to the "Computer Science" department.

Solution: - db.enrollment.aggregate([{$lookup: {from: "students",localField:"student_id",foreignField: "_id",as: "student" }},{ $unwind: "$student" },{$lookup: {from: "courses",localField: "course_id",foreignField: "_id",as: "course"}},{ $unwind: "$course" },{$match: {"marks": { $gt: 80 },"course.title": "Web Development","student.department": "Computer Science"}},{$project: { _id: 0,student_name: "$student.name",course_title: "$course.title",marks: 1,department: "$student.department"}}])

Explanation: -

- **$lookup →** Joins enrollments with students_full and courses_full.
- **$match →** Filters students with marks >80 in "Web Development" and in "Computer Science".
- **$project →** Shows student name, course title, marks, and department.

Output:



Q13. For each faculty member, list the names of all students enrolled in their courses along with average marks per student per faculty.

Solution: db.faculty.aggregate([{$lookup:{from:"courses",localField:"_id",foreignField:"faculty_id",as: "courses"}},{$unwind:"$courses"},{$lookup:{from:"enrollment",localField:"courses._id",foreignField: "course_id",as: "enrollment" }},{ $unwind: "$enrollment" },{$lookup: {from: "students",localField: "enrollment.student_id", foreignField:"_id", as: "student"}},{ $unwind: "$student" },{$group: { _id: { faculty: "$name", student: "$student.name" }, avg_marks: { $avg: "$enrollment.marks" }}}, {$group: { _id:"$_id.faculty",students: { $push: { name: "$_id.student",average_marks: "$avg_marks" }}}},{$project: {_id: 0,faculty_name: "$_id", students: 1}}])

Explanation: -

- **Joins courses_full → enrollments_full → students_full.**
- **$group →** Groups by faculty ID and collects all student names and their average marks.
- **$lookup →** Fetches faculty names.
- **$round →** Rounds average marks to 2 decimals.

Output: -

```
db> db.faculty.aggregate([{$lookup:{from:"courses",localField:"_id",foreignField:"faculty_id",as: "courses"}},{$unwind:"$courses"},{$lookup:{from:"enrollmen
t",localField:"courses._id",foreignField: "course_id",as: "enrollment" }},{ $unwind: "$enrollment" },{$lookup: {from: "students",localField: "enrollment.stu
dent_id", foreignField:"_id", as: "student"}},{ $unwind: "$student" },{$group: { _id: { faculty: "$name", student: "$student.name" }, avg_marks: { $avg: "$e
nrollment.marks" }}}, {$group: { _id:"$_id.faculty",students: { $push: {  name: "$_id.student",average_marks: "$avg_marks" }}}},{$project: {_id: 0,faculty_n
faculty_name: "$_id", students: 1}}]) //Pratishtha Srivastava 1240258328
[
  {
    students: [
      { name: 'Carolyn Chandler', average_marks: 51 },
      { name: 'Logan Murphy', average_marks: 54 },
      { name: 'Rachel Maldonado', average_marks: 71 }
    ],
    faculty_name: 'Michael Johnson'
  },
  {
```

Q14. Show the most popular activity type (e.g., Hackathon, Seminar, etc.) by number of student participants.

Solution: - db.activities.aggregate([{ $group: { _id: "$type", participants: { $addToSet: "$student_id" } } }, { $project: { _id: 0, activity_type: "$_id", number_of_participants: { $size: "$participants" } } }, { $sort: { number_of_participants: -1 } }, { $limit: 1 }])

Explanation: -

- **$group →** Groups activities by type and collects unique student IDs.
- **$size →** Counts number of participants per activity type.
- **$sort →** Orders in descending order.
- **$limit: 1 →** Returns most popular activity.

Output: -

```
Type "it" for more
db> db.activities.aggregate([ { $group: { _id: "$type", participants: { $addToSet: "$student_id" } } }, { $project: { _id: 0, activity_type: "$_id", number_
of_participants: { $size: "$participants" } } }, { $sort: { number_of_participants: -1 } }, { $limit: 1 } ]) //Pratishtha Srivastava 1240258328
[ { activity_type: 'Hackathon', number_of_participants: 29 } ]
db>
```